1. The site of protein synthesis in a cell is

A) Nucleus

B) Ribosome

C) Mitochondria

D) Golgi body

Answer: B

---

2. The fluid mosaic model of cell membrane was proposed by

A) Watson and Crick

B) Singer and Nicolson

C) Mendel

D) Darwin

Answer: B

---

3. The process by which water moves across a semipermeable membrane is

A) Diffusion

B) Osmosis

C) Active transport

D) Transpiration

Answer: B

---

4. Which phase of mitosis involves alignment of chromosomes at the equator?

A) Prophase

B) Metaphase

C) Anaphase

D) Telophase

Answer: B

---

5. The enzyme responsible for DNA replication is

A) RNA polymerase

B) DNA polymerase

C) Ligase

D) Helicase

Answer: B

---

6. Which organelle modifies and packages proteins?

A) Endoplasmic reticulum

B) Golgi apparatus

C) Ribosome

D) Lysosome

Answer: B

---

7. The genetic material in most organisms is

A) RNA

B) DNA

C) Protein

D) Lipid

Answer: B

---

8. The number of chromosomes in a human gamete is

A) 46

B) 23

C) 44

D) 22

Answer: B

---

9. Crossing over occurs during

A) Prophase I

B) Metaphase I

C) Anaphase I

D) Telophase I

Answer: A

10. The main function of mitochondria is

A) Protein synthesis

B) Lipid synthesis

C) ATP production

D) Digestion

Answer: C

---

11. The pigment responsible for photosynthesis is

A) Hemoglobin

B) Chlorophyll

C) Melanin

D) Carotene

Answer: B

---

12. The light reaction of photosynthesis occurs in

A) Stroma

B) Thylakoid

C) Cytoplasm

D) Nucleus

Answer: B

---

13. The dark reaction occurs in

A) Thylakoid

B) Stroma

C) Cytoplasm

D) Nucleus

Answer: B

---

14. The opening and closing of stomata is controlled by

A) Guard cells

B) Xylem

C) Phloem

D) Epidermis

Answer: A

---

15. The process of loss of water in plants is

A) Transpiration

B) Respiration

C) Photosynthesis

D) Guttation

Answer: A

---

16. The transport of water in plants occurs through

A) Phloem

B) Xylem

C) Cortex

D) Epidermis

Answer: B

---

17. Which hormone promotes cell division?

A) Auxin

B) Cytokinin

C) Ethylene

D) Abscisic acid

Answer: B

---

18. The structural unit of kidney is

A) Neuron

B) Nephron

C) Glomerulus

D) Tubule

Answer: B

---

19. The oxygen-carrying pigment is

A) Chlorophyll

B) Hemoglobin

C) Myoglobin

D) Melanin

Answer: B

---

20. The cardiac muscle is

A) Voluntary

B) Involuntary

C) Striated

D) Both B and C

Answer: D

---

21. The normal pH of human blood is

A) 6.8

B) 7.4

C) 8.0

D) 7.0

Answer: B

---

22. The largest lymphoid organ is

A) Liver

B) Spleen

C) Thymus

D) Kidney

Answer: B

---

23. The hormone insulin regulates

A) Protein

B) Fat

C) Blood sugar

D) Water

Answer: C

---

24. The part of brain responsible for balance is

A) Cerebrum

B) Cerebellum

C) Medulla

D) Hypothalamus

Answer: B

---

25. Reflex action is controlled by

A) Brain

B) Spinal cord

C) Cerebellum

D) Hypothalamus

Answer: B

---

26. The functional unit of nervous system is

A) Neuron

B) Axon

C) Synapse

D) Dendrite

Answer: A

---

27. The site of fertilization in humans is

A) Uterus

B) Ovary

C) Fallopian tube

D) Cervix

Answer: C

---

28. The hormone responsible for ovulation is

A) Estrogen

B) Progesterone

C) LH

D) FSH

Answer: C

---

29. The male gamete is

A) Ovum

B) Sperm

C) Zygote

D) Embryo

Answer: B

---

30. The female gamete is

A) Ovum

B) Sperm

C) Zygote

D) Embryo

Answer: A

---

31. The theory of evolution was given by

A) Mendel

B) Darwin

C) Lamarck

D) Watson

Answer: B

---

32. Natural selection was proposed by

A) Darwin

B) Mendel

C) Lamarck

D) Morgan

Answer: A

---

33. The unit of classification is

A) Species

B) Genus

C) Family

D) Order

Answer: A

---

34. The scientific name consists of

A) One word

B) Two words

C) Three words

D) Four words

Answer: B

---

35. The study of fungi is

A) Botany

B) Mycology

C) Zoology

D) Ecology

Answer: B

---

36. The study of bacteria is

A) Virology

B) Bacteriology

C) Mycology

D) Cytology

Answer: B

---

37. The virus contains

A) DNA only

B) RNA only

C) Both DNA or RNA

D) Protein only

Answer: C

---

38. Vaccines provide

A) Passive immunity

B) Active immunity

C) Natural immunity

D) Artificial immunity

Answer: B

---

39. AIDS is caused by

A) Bacteria

B) Virus

C) Fungus

D) Protozoa

Answer: B

---

40. Malaria is caused by

A) Virus

B) Bacteria

C) Protozoa

D) Fungus

Answer: C

41. The causative agent of tuberculosis is

A) Virus

B) Bacteria

C) Fungus

D) Protozoa

Answer: B

---

42. Antibiotics are effective against

A) Viruses

B) Bacteria

C) Fungi

D) Protozoa

Answer: B

---

43. The primary source of energy is

A) Protein

B) Fat

C) Carbohydrates

D) Vitamins

Answer: C

---

44. Enzymes are

A) Carbohydrates

B) Proteins

C) Lipids

D) Vitamins

Answer: B

---

45. The deficiency of iodine causes

A) Anemia

B) Goitre

C) Rickets

D) Scurvy

Answer: B

---

46. Vitamin D deficiency causes

A) Scurvy

B) Rickets

C) Night blindness

D) Beriberi

Answer: B

---

47. The process of formation of urine is

A) Filtration

B) Reabsorption

C) Secretion

D) All

Answer: D

---

48. The lungs are responsible for

A) Digestion

B) Respiration

C) Circulation

D) Excretion

Answer: B

---

49. The structure of DNA is

A) Single helix

B) Double helix

C) Triple helix

D) Linear

Answer: B

---

50. RNA differs from DNA in having

A) Thymine

B) Uracil

C) Adenine

D) Guanine

Answer: B

---

51. The central dogma is

A) DNA → RNA → Protein

B) RNA → DNA → Protein

C) Protein → DNA → RNA

D) DNA → Protein → RNA

Answer: A

---

52. The number of autosomes in humans is

A) 22 pairs

B) 23 pairs

C) 44 pairs

D) 46 pairs

Answer: A

---

53. The sex chromosomes in males are

A) XX

B) XY

C) YY

D) XO

Answer: B

---

54. Mutation is

A) Normal change

B) Sudden genetic change

C) Growth

D) Reproduction

Answer: B

---

55. Biotechnology deals with

A) Animals

B) Plants

C) Use of organisms

D) Cells

Answer: C

---

56. Genetic engineering involves

A) DNA manipulation

B) Cell division

C) Respiration

D) Digestion

Answer: A

---

57. Cloning produces

A) Variation

B) Identical copies

C) Mutations

D) Hybrids

Answer: B

---

58. Ecosystem consists of

A) Living only

B) Non-living only

C) Both

D) None

Answer: C

---

59. The flow of energy in ecosystem is

A) Cyclic

B) Unidirectional

C) Random

D) Infinite

Answer: B

---

60. The ultimate source of energy is

A) Moon

B) Sun

C) Earth

D) Wind

Answer: B